Exp No:- 02 DETERMINAION OF CURRENT IN
DATE :- 03/02/25 CIRCUIT USING MESH ANALYSIS

AIM :-

To Determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit

APPARATUS REQUIRED :-

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power supply (RPS)	(0-30V)	1
2.	Ammeter	(0-10mA) MC	3
3.	Resistors	330Ω, 470Ω, 560Ω	Each 1
4.	Bread Board	---	1

PROCEDURE :-

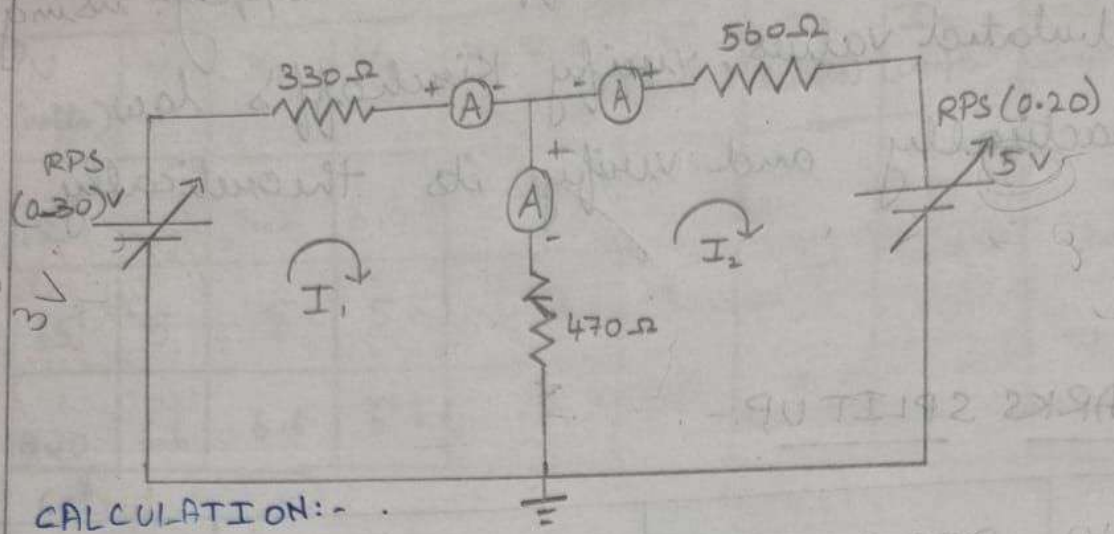
1. Give connections as per the circuit Diagram.
2. Switch ON the supply, vary the RPS (Regulated Power supply) and set a particular input voltage.
3. Note down the readings of ammeters and voltmeters and tabulate them.

TABULAR COLUMN:-

	Parameters	Theoretical	Practical
1.	330 (I_1)	1.2	1.3
2.	470 (I_2)	4.2	4.3
3.	560 (I_1+I_2)	5.4	5.4

o/p verified

CIRCUIT DIAGRAM :-



CALCULATION:-

$$800 I_1 - 470 I_2 = 3 \rightarrow (1)$$

$$-470 I_1 + 1030 I_2 = 5 \rightarrow (2)$$

$$\Delta = \begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Delta = 1030.$$

$$\Delta_1 = \begin{bmatrix} 3 & -470 \\ -5 & 1030 \end{bmatrix} = 3 \times 1030 - (-470 \times -5)$$

$$= 3090 - 2350$$

$$= 740$$

$$\Delta_2 = \begin{bmatrix} 800 & 3 \\ -470 & -5 \end{bmatrix} = 800 \times -5 - (-470 \times 3)$$

$$= -4000 + 1410$$

$$= 2590.$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{1030} = 1.226.$$

$$I_2 = \frac{2590}{1030} = 4.294$$

$$I_3 = I_1 + I_2 = 1.226 + 4.294$$

$$\boxed{I_3 = 5.52}$$

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4. Vary the RPS for different input voltages and note down the readings of all the meters.

5. Reduce the RPS to its minimum value and switch OFF the supply.

6. using the tabulated values, verify Kirchoff's laws practically, and verify it theoretically.

MARK SPLIT UP:-

SL.NO	DESCRIPTION	MARK ALLOTED	MARK OBTAINED
1.	Experimental setup / connections.	20	20
2.	observation / Readings.	20	20
3.	calculations / Result.	20	20
4.	Viva.	20.	20
5.	Record.	~~Total~~ 20	20
		Total	100

100
4/2/26

RESULT:-

Thus, the MESH Analysis is verified practically & theoretically. The resultant currents for __1__ V supply are:

a) The current across 330 Ω resistor is 1.226

b) The current across 560 Ω resistor is 4.294.

c) The current across 470 Ω resistor is 5.52.